

Problem

Determine the total charge transferred over the time interval of $0 \leq t \leq 10$ s when

$$i(t) = \frac{1}{2}t \text{ A}.$$

Step-by-step solution

Step 1 of 1

Consider the following data:

The expression for current $i(t)$ is as follows.

$$i(t) = \frac{1}{2}t \text{ A}$$

Calculate the total charge transferred over the time interval of $0 \leq t \leq 10$ s by using following formula.

$$Q \triangleq \int_{t_0}^t i \, dt$$

Substitute 0 for t_0 , 10 for t and $\frac{1}{2}t \text{ A}$ for $i(t)$.

$$\begin{aligned} Q &\triangleq \int_0^{10} \left(\frac{1}{2}t \text{ A} \right) dt \\ &\triangleq \frac{1}{2} \int_0^{10} (t \text{ A}) dt \\ &\triangleq \frac{1}{2} \left(\frac{t^2}{2} \right) \bigg|_0^{10} \\ &\triangleq \frac{1}{2} \left(\frac{(10)^2}{2} - \frac{(0)^2}{2} \right) \\ &\triangleq \frac{1}{2} \left(\frac{10000}{2} - 0 \right) \\ &\triangleq \frac{100}{4} \\ &\triangleq 25 \text{ C} \end{aligned}$$

Therefore, the total charge transferred over the time interval of $0 \leq t \leq 10$ s is 25C.